1.7.3 ▶ ▶ ▶ Continuity Equation

The continuity equation represents the application of conservation of mass to fluids. Consider fluid flowing ideally through a section of a pipe with cross-section A_1 on the left side and A_2 on the right. The fluid flows at a speed v_1 on the side with area A_1 and vice versa. In a certain time Δt, a certain volume of fluid V_1 enters the tube on the left and a certain volume V_2 exits the tube on the right. The volume of water entering the tube is given by

$$V_1 = A_1 v_1 \Delta t$$

and the volume of water exiting the tube is given by

$$V_2 = A_2 v_2 \Delta t.$$

Since the fluid is incompressible and mass is conserved, the amount of fluid flowing in must be equal to the volume of water flowing out. Thus,

$$A_1 v_1 \Delta t$$

$$\boxed{A_1 v_1 = A_2 v_2} \tag{1.113}$$

The above equation is known as the continuity equation, and is the manifestation of conservation of mass in the flow of ideal fluids.

▶ For a more general equation of continuity that extends beyond fluid flow in a pipe, consider an arbitrary volume element with volume V and boundary dV. The net flux of fluid into the volume element is given by

$$\Phi_{net} = \oiint_{dV} \mathbf{u} \cdot d\mathbf{A}$$

where Φ_{net} is the net volume flow rate of fluid, $\mathbf{u}$ is the velocity field of the fluid, and $d\mathbf{A}$ is an area element on the surface dV. Since the fluid is incompressible and mass isn't created or destroyed, the net flux through the volume must be zero.

$$\oiint_{dV} \mathbf{u} \cdot d\mathbf{A} = 0 \tag{1.114}$$

Using the divergence theorem, we can alternatively write eq. 1.104 as

$$\boxed{\nabla \cdot \mathbf{u} = 0}. \tag{1.115}$$

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